

Domenico Valentino

Montreal, QC | 514-977-7505 | domenico2727@icloud.com
github.com/domilx | LinkedIn

EDUCATION

McGill University | B.Eng., Computer Engineering | Expected May 2030

Dawson College | Mathematics and Computer Science | Honour Roll | May 2026

TECHNICAL SKILLS

Languages: C/C++, Java, Rust, Python, 6502 assembly

Controls and hardware: WPILib, PID/feedforward, sensor integration, digital circuits

Tools and systems: Linux, Git, PathPlanner, SysId, NetworkTables, audio DSP, concurrency

ENGINEERING EXPERIENCE

FIRST Robotics | Programming Lead and Robotics Mentor | 2022 - 2025

- Authored and tested competition-deployed Java/WPILib robot software, integrating a four-module swerve drivetrain, PathPlanner autonomous driving, and scoring mechanisms.
- Integrated wheel encoders, a gyro, and three AprilTag cameras through WPILib's pose estimator; adjusted vision measurement uncertainty using tag distance and count.
- Developed vision-guided PID alignment, drivetrain/elevator SysId routines, and NetworkTables telemetry for control tuning and debugging.
- Led 10 programmers across teams 3990 and 9406; delegated tasks, reviewed pull requests, and taught beginners who progressed to writing robot control systems.

Team 3990 award (2024): Long Island Autonomous Award.

2025 Championship match video

Kaskaraa Instruments | Co-Founder and Software Lead | Apr 2025 - Present

kaskaraa.com | District 3 incubation | Mitacs-supported research

- Lead software development for prototype automated microtomes that prepare thin tissue sections for pathology, including UI, motor control, safety logic, and AI imaging.
- Develop custom embedded platforms, circuit boards, and wiring, integrating electronics with instrument software.
- Manage interns and collaborate with ~10 Concordia graduate engineering students.

ENGINEERING PROJECTS

6502 Computer | Digital hardware and assembly | Summer 2026

- Built and programmed a working 6502 computer; wired CPU, memory, clock/reset, address decoding, and I/O, then brought up the system using ROM programming and assembly.

OverCue | DJ software and firmware | Jul 2026 - Present

overcue.gg | Publicly released software and firmware

- Developed C/C++ audio extensions for ARM64 Linux music players, using atomic synchronization, bounded buffers, resampling, and click-suppressed playback transitions.
- Added DJM knob control of CDJ stems with switching back to standard EQ; external apps handle stem processing.
- Tested audio processing for concurrency, cancellation, malformed inputs, and loudness limits.

ADDITIONAL EXPERIENCE

Apple | Product Specialist | Oct 2025 - Present

- Explain product capabilities and system behavior in plain language; serve as a technical resource for colleagues troubleshooting customer questions.
 - De-escalate difficult customer interactions by explaining product behavior and the reasons behind recommendations.
- English and French (fluent); Italian and Spanish (conversational).